

Networking Basics



Networking Basics & IP Addressing

▪ **Networking Basics**

- OSI TCI/IP Model
- Port Number
- Network Architecture
- Wireshark

▪ **IP Addressing**

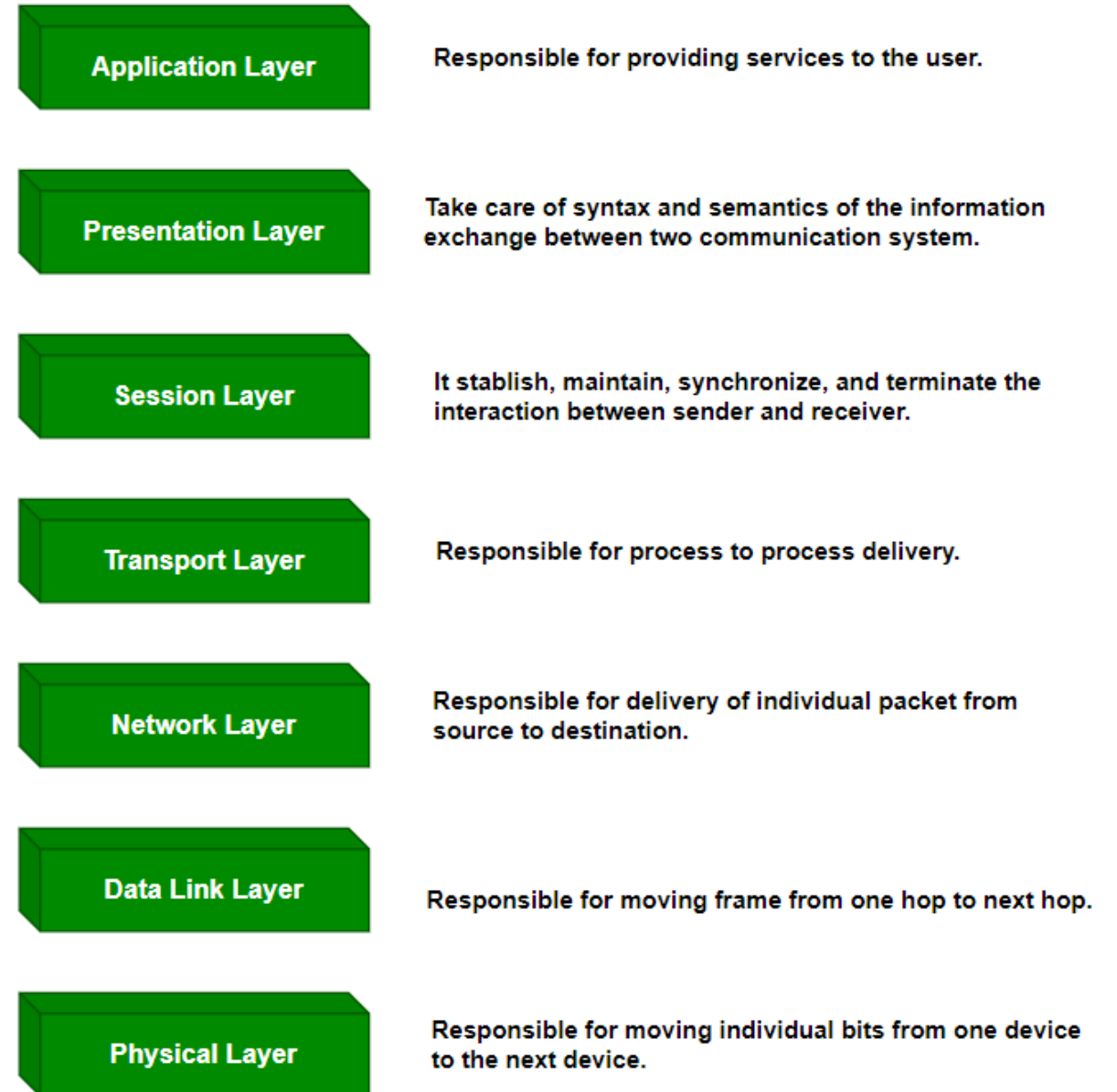
- Introduction to IP Addressing
- IPv4 Addressing
- Network Addressing
- Classless Inter-Domain Routing
- Private IP Address
- APIPA
- IPv6 Addressing
- Network Address Translation
- Subnet and Super net

Labs

- Students will configure IP address in Windows machine and in Kali linux.
- Ensure that they are in the same network and also ping each other.
- Students will practice different networking commands.
- They will build a network of machines using the concept of docker and use different commands.

OSI Layer

- Open System Interconnect (OSI) model.
- OSI is used to define *how data is sent from one computer to another through network*.
- Introduced by International Organization for Standardization (ISO) in 1984.
- It contains 7 layers.



Wanna remember this?

All

People

Should

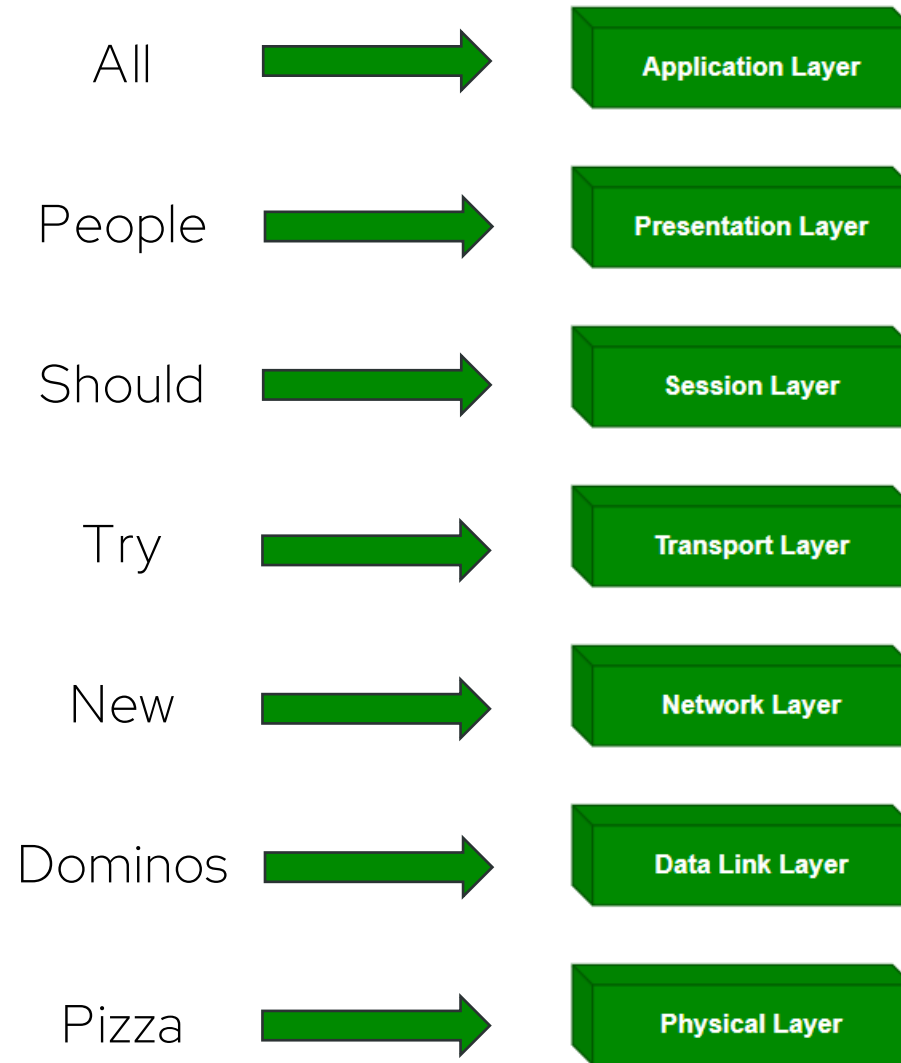
Try

New

Dominos

Pizza

OSI Layer



OSI Layers

- **Application Layer:** Applications create the data.
- **Presentation Layer:** Data is formatted, compressed and encrypted.
- **Session Layer:** Connections are established and managed.
- **Transport Layer:** Data is broken into segments for reliable delivery.
- **Network Layer:** Segments are packaged into packets and routed.
- **Data Link Layer:** Packets are framed and sent to the next device.
- **Physical Layer:** Frames are converted into bits and transmitted physically.

Application Layer

Presentation Layer

Session Layer

Transport Layer

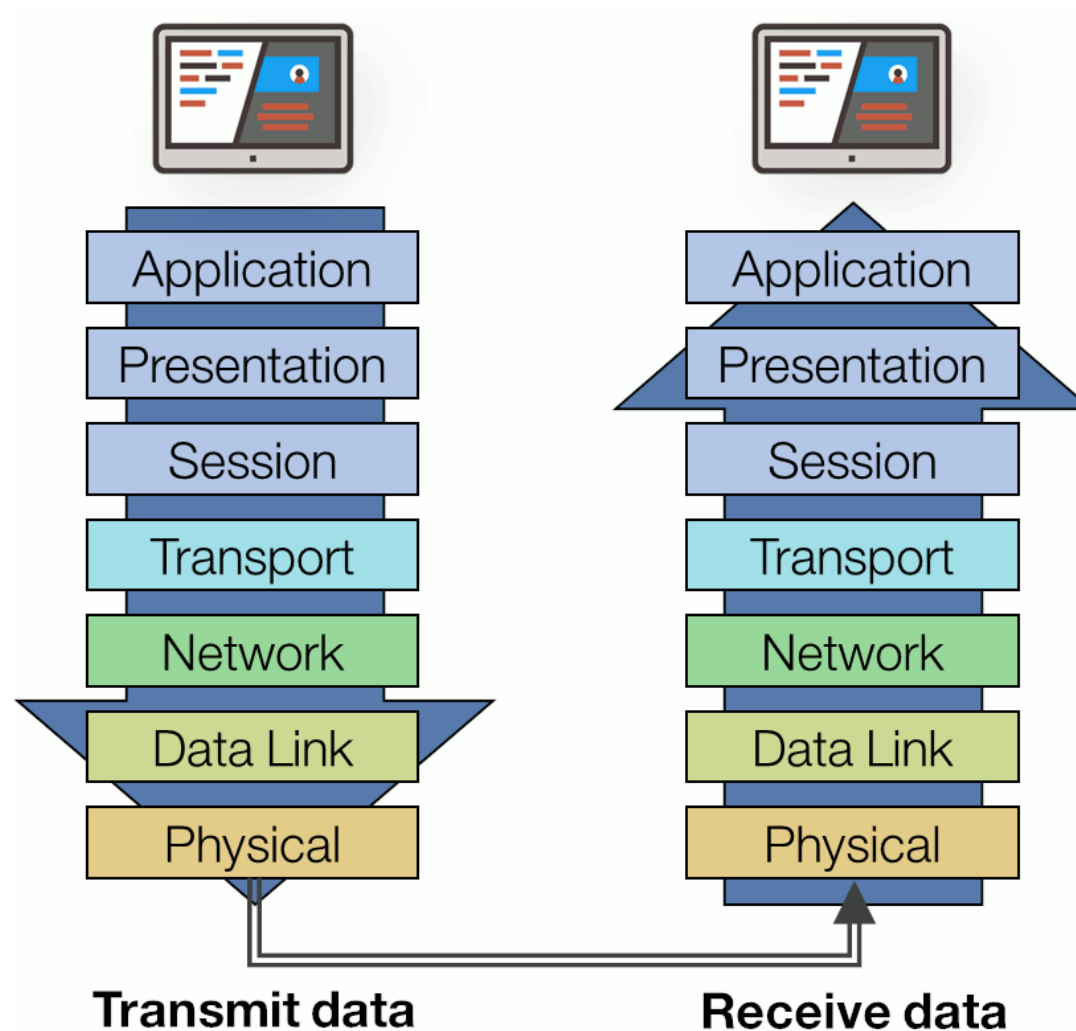
Network Layer

Data Link Layer

Physical Layer



OSI Layers – Sender & receiver



Layer 7: Application layer

- The Application layer is the **highest layer** of the OSI model.
- It provides the interface between the **network protocol** and the **software running** on the computer.
- This layer is used by “**networking applications**” like:
 - Google Chrome, Firefox, Skype, Teams, Etc....
- It facilitates **data formatting** and **translation for application** use.
- It uses **common protocols** like:
 - HTTP, FTP, SMTP, DNS, Telnet, SSH, IMAP, POP, SNMP, etc.



Layer 6: presentation layer

- This layer receives data from application layer in form of **characters & numbers**.
- This layer converts these characters into **binary format** & is called “**translation**”.
- Here, **data is compressed** which requires less size to transmit which could be useful in scenarios like video streaming.
- It also **encrypts the data** by using SSL for secured transmission.
- It supports protocols like: **SSL/TLS, JPEG, MPEG, ASCII**
- It performs:
 - **Data translation** (into binary)
 - **Data encryption/decryption** (for security)
 - **Data compression** (for reduced size)



Layer 5: Session layer

- The session layer **manages sessions** or **connections** between applications.
- It **establishes, manages, and terminates sessions**, ensuring that the data exchange between systems is synchronized.
- This layer is responsible for:
 - **Authentication** (who you are?)
 - **Authorization** (do you have permission?)
 - **Placement of header** information in a packet (where a message starts and where it ends)
- It supports protocols like: NetBIOS, RPC, SOCKS, L2TP, SDP, H.245, NFS, etc.
- Controls whether the data being exchanged in a session are transmitted as **full or half duplex** messages.



Layer 4: Transport layer

- The Transport Layer ensures that messages are delivered **error-free, in sequence** and with **no loss or duplication**.
- It controls reliability of communication through:
 - **Segmentation** (divides data into small data units called segments)
 - **Flow control** (controls amount of data flow)
 - **Error control** (Automatic repeat request in case of transmission lost)
- This layer has:
 - Transmission Control Protocol (**TCP**) – **Connection Oriented** transmission
 - User Datagram Protocol (**UDP**) – **Connection-less** transmission



TCP vs UDP

UDP	TCP
- No feedback	- Send feedback
- Faster than TCP	- Slower in nature
- Example:	- Example:
- DNS	- World Wide Web
- Online games	- File Transfer Protocol (FTP)
- Online video streaming	- Emails
- Radio	



LAYER 3: NETWORK LAYER (INTERNET LAYER)

- Transport layer sends **segments** to this layer. Here, data units are called “**Packets**”.
- This layer is primarily responsible for establishing the paths used for **transfer of data packets** between nodes on the network. This is the layer that **routers** operate on.
- **Path determination** (finding best path of delivery).
- This layer is responsible for:
 - **Logical addressing** (IP addresses)
 - **Packet forwarding** (routing through different routers)
 - **Fragmentation** and reassembly of packets
 - Handling **traffic control** (congestion management)



Layer 3: Network Layer (Internet Layer)

- The network layer also takes care of **mapping logical (IP) addresses to physical (MAC) addresses** that are used in the Data Link layer.
- Supported protocols:
 - **IP** (Internet Protocol) - IPv4 & IPv6
 - **ICMP** (Internet Control Message Protocol)
 - **RIP** (Routing Information Protocol)
 - **OSPF** (Open Shortest Path First)
 - **BGP** (Border Gateway Protocol)
 - **MPLS** (Multiprotocol Label Switching)



Layer 2: Data Link Layer

- At the Data Link layer, data packets are encoded into **bits**.
- This layer is responsible for **establishing, maintaining, and terminating** a link between two directly connected nodes.
- It ensures **error-free** transmission between devices on the same network and controls how data is placed on the physical medium.
- MAC address of sender & receiver is assigned to packet to create a frame.
 - MAC address is a **12 digit, alpha-numeric number** embedded on NIC by manufacturer.
- Supported protocols: **ARP, PPP, Token ring, L2TP**, etc



Layer 1: Physical Layer

- This layer converts data from data link layer into signals.
- This layer is the lowest layer of the OSI model and consists of the functionality that interacts with the actual hardware and signaling mechanism.
- This layer is responsible for:
 - Defines the **physical characteristics** of the network (cables, connectors, voltage levels).
 - Handles the **transmission of binary data** (0s and 1s) as electrical, optical, or radio signals.
 - Manages the **data rate and synchronization** of bit-level communication.
- Supported protocols: Ethernet, USB, Bluetooth, Infrared,



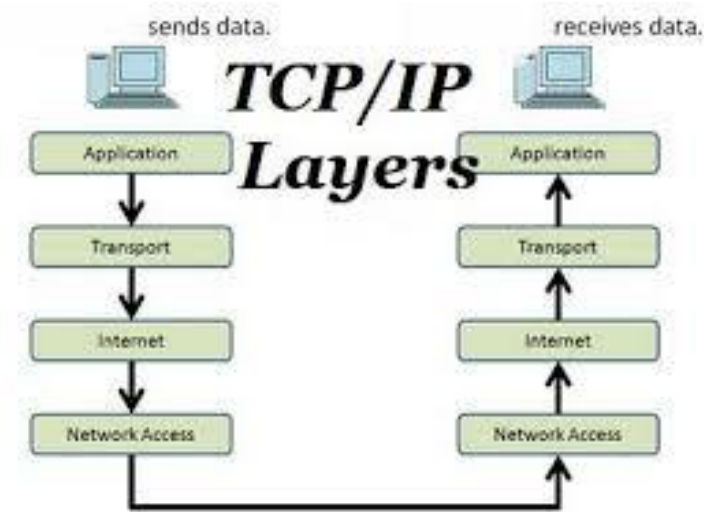
TCP/IP Model

OSI Model	TCP/IP (4 layer)	TCP/IP (5 layer)
Application	Application	Application
Presentation		
Session		
Transport	Transport	Transport
Network	Network	Internet layer
Data Link	Network Access	Data link
Physical		Physical layer



TCP/IP Model - Application Layer

- The TCP/IP model is a fundamental framework for computer networking.
- It consists of four layers: the Link Layer, the Internet Layer, the Transport Layer, and the Application Layer.
- Each layer has specific functions that help manage different aspects of network communication, making it essential for understanding and working with modern networks.
- The main work of TCP/IP is to transfer the data of a computer from one device to another.
- The TCP/IP model is used in the context of the real-world internet, where a wide range of physical media and network technologies are in use.



TCP/IP Model - Application Layer

- This model defines how data is transmitted over networks, ensuring reliable communication between devices.
- This layer corresponds to the Application, Presentation, and Session layers of the OSI model.
- This layer:
 - **Provides protocols** that allow software applications to send and receive data over the network.
 - Manages the **formatting, encryption, and compression of data** for use by applications.
 - Supports the **establishment, maintenance, and termination of communication** sessions between applications.



TCP/IP - Application Layer

- This protocol supports:
 - **HTTP/HTTPS** (Hypertext Transfer Protocol/Secure) - Web browsing.
 - **FTP** (File Transfer Protocol) - File transfers.
 - **SMTP** (Simple Mail Transfer Protocol) - Email transmission.
 - **DNS** (Domain Name System) - Resolving domain names to IP addresses.
 - **Telnet**, SSH (Secure Shell) - Remote access to servers.
 - **SNMP** (Simple Network Management Protocol) - Network management.

TCP/IP - Transport Layer

- The Transport Layer ensures reliable **data transfer** between devices.
- It is responsible for maintaining the **end-to-end communication, error-checking, and data flow control**.
- Functionalities of this layer:
 - **Establishes, maintains, and terminates** connections between devices.
 - **Segments** and **reassembles** data into a format that can be transmitted over the network.
 - **Provides flow control to prevent network congestion** and ensures that data is delivered in sequence.
 - Offers error-checking mechanisms to **detect and recover** from data transmission errors.



TCP/IP - Transport Layer

- Standard protocols used here are:
 - TCP (**Transmission Control Protocol**) - Ensures reliable, connection-oriented communication.
 - UDP (**User Datagram Protocol**) - Supports fast, connectionless communication with no guarantee of delivery.



TCP/IP - Internet Layer

- The Internet Layer is responsible for routing data across networks and ensuring that it reaches its destination.
- It is equivalent to the Network Layer in the OSI model.
- In this layer:
 - **Logical addressing** (IP addresses) to identify devices across different networks.
 - **Routing of data packets** from the source network to the destination network, possibly across multiple networks.
 - **Fragmentation and reassembly** of data packets if the data exceeds the size limit of a network.
- This protocol supports: **IP, ICMP, IGMP, IPSec**



TCP/IP - Network Interface Layer (Link Layer)

- The Network Interface Layer is responsible for the physical transmission of data over a network. It corresponds to the combination of the Physical and Data Link layers in the OSI model.
- In this layer, it:
 - Defines how data is sent over the physical medium (cables, radio waves).
 - Handles hardware addressing (MAC addresses) and error detection within the local network.
 - Manages frame placement on the network and ensures data reaches its immediate destination within the same network.



TCP/IP - Network Interface Layer (Link Layer)

- This layer supported protocols/standards:
 - Ethernet
 - Wi-Fi (IEEE 802.11)
 - ARP (Address Resolution Protocol)
 - PPP (Point-to-Point Protocol)

OSI – TCP/IP in a nutshell

TCP/IP	OSI Layer	Functions	PDU's	Protocols	Devices
Application Layer	Application	User layer	Data	HTTP, FTP, SMTP	App FW, VPN, SSL, Proxies
	Presentation	Encrypt, Compress, Converts		ASCII, JPEG, DOC	
	Session	Open & Closes session		RPC, TLS	
Transport	Transport	Ensures Delivery	Segments	TCP, UDP	LB, N/W Firewall
Network	Network	Control Routing	Packet	IP, IPSec	Routers, L3
Physical Layer	Data Link	Error free data transfer	Frame	Ethernet, MAC	NIC, Switches
	Physical Layer	BITS			RJ45

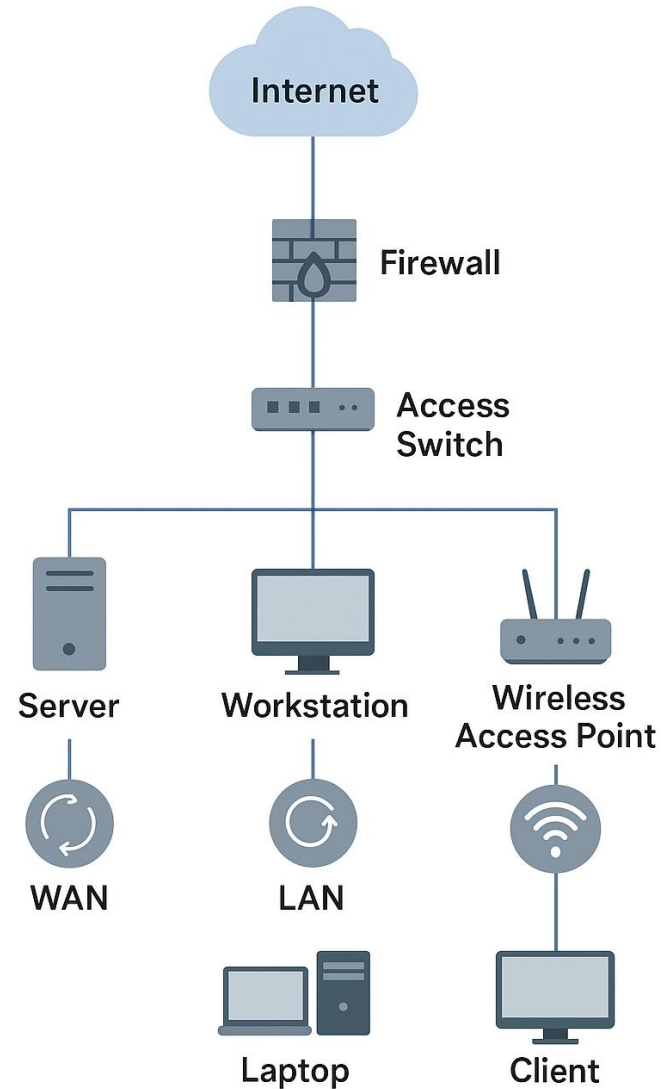


Port Numbers

Port	Protocol	Usage / Description
20/21	FTP	File Transfer Protocol
22	SSH	Secure remote login, SCP, SFTP
23	Telnet	Unsecured remote login (legacy)
25	SMTP	Sending email (mail transfer)
53	DNS	Domain Name System (queries & zone transfers)
67/68	DHCP	Dynamic Host Configuration Protocol (IP assignment)
69	TFTP	Trivial FTP (simple file transfer)
80	HTTP	Web traffic (unencrypted)
110	POP3	Receiving email (post office protocol)
123	NTP	Network Time Protocol (time sync)
143	IMAP	Email retrieval (access on server)
161/162	SNMP	Network monitoring (queries/traps)
389	LDAP	Directory services (Active Directory)
443	HTTPS	Secure web traffic (TLS/SSL)
445	SMB	Windows file sharing (modern)
514	Syslog	System log messages
587	SMTP (TLS)	Secure mail submission
1433	MS SQL	Microsoft SQL Server
1521	Oracle DB	Oracle database service
2049	NFS	Network File System
3306	MySQL	MySQL database
3389	RDP	Remote Desktop Protocol



Network Architecture



Wireshark

Wireshark is a *free, open-source packet analyzer* used to inspect network traffic at the most granular level.

It captures packets in real time and displays them for detailed inspection, making it one of the most essential tools for:

- Network troubleshooting
- Performance analysis
- Security investigations
- Protocol research and learning



Key Features of Wireshark

- **Real-time Packet Capture**

- Captures live traffic from network interfaces (Ethernet, Wi-Fi, VPN, etc.)
- Displays each packet with timestamps, addresses, protocols, and payloads.

- **Deep Inspection of Over 3,000 Protocols**

- HTTP, HTTPS, DNS, DHCP, TCP/UDP, SSL/TLS, SMB, and many more..

- **Color Coding (Display Filters)**

- Color rules help identify different types of traffic:
 - TCP errors (red)
 - DNS queries (blue)
 - HTTP traffic (green)



Key Features of Wireshark

- **Powerful Filtering**

- http – show only HTTP packets
- ip.addr == 10.1.1.1 – filter packets from/to an IP
- tcp.port == 443 – show HTTPS packets
- dns – display DNS queries and responses

- **Packet Reassembly**

- Reassemble:
 - TCP streams
 - SSL sessions (if keys available)
 - VoIP streams

- **VoIP Analysis**

- Decode SIP, RTP, and audio streams
- Playback captured audio

How Wireshark Works ?

Wireshark has three main components:

- **Capture Engine (Dumpcap)**

- Handles packet capture
- Uses OS-level libraries: WinPcap/Npcap, libpcap

- **Dissection Engine**

- Converts raw packet data into human-readable protocol fields

- **GUI + Analysis Tools**

- Displays packet list, detailed view, protocol hierarchy, graphs

Packet Capture Process

1. Wireshark starts capturing on a NIC
2. NIC enters promiscuous mode
3. Packets from the network are passed to Wireshark
4. Wireshark decodes the packet using protocol dissectors
5. Displays structured details to the user



192.168.1.8

No.	Time	Source	Destination	Protocol	Length	Info
5780	155.333994	64.233.170.84	192.168.1.8	QUIC	63	Protected Payload (KP0)
5781	155.334230	192.168.1.8	64.233.170.84	QUIC	73	Protected Payload (KP0), DCID=fbef5a2d2ac6e38f
5782	155.334344	192.168.1.8	64.233.170.84	QUIC	77	Protected Payload (KP0), DCID=fbef5a2d2ac6e38f
5783	155.359764	192.168.1.8	64.233.170.84	QUIC	74	Protected Payload (KP0), DCID=fbef5a2d2ac6e38f
5784	155.430905	64.233.170.84	192.168.1.8	QUIC	66	Protected Payload (KP0)
5785	155.532225	192.168.1.8	192.168.1.101	TLSv1.2	164	Application Data
5786	155.532225	192.168.1.8	192.168.1.101	TLSv1.2	164	Application Data
5787	155.532300	192.168.1.8	192.168.1.101	TLSv1.2	164	Application Data
5788	155.650867	192.168.1.101	192.168.1.8	TLSv1.2	164	Application Data
5789	155.650867	192.168.1.101	192.168.1.8	TLSv1.2	164	Application Data
5790	155.650867	192.168.1.101	192.168.1.8	TLSv1.2	164	Application Data
5791	155.705088	192.168.1.8	192.168.1.101	TCP	54	50733 → 8009 [ACK] Seq=3411 Ack=3411 Win=255 Len=0
5792	155.705088	192.168.1.8	192.168.1.101	TCP	54	52248 → 8009 [ACK] Seq=3411 Ack=3411 Win=250 Len=0
5793	155.705088	192.168.1.8	192.168.1.101	TCP	54	60637 → 8009 [ACK] Seq=3411 Ack=3411 Win=252 Len=0
5794	156.146819	192.168.1.8	192.168.1.102	TLSv1.2	164	Application Data
5795	156.151724	192.168.1.102	192.168.1.8	TLSv1.2	164	Application Data
5796	156.193715	192.168.1.8	192.168.1.102	TCP	54	52247 → 8009 [ACK] Seq=3631 Ack=3639 Win=252 Len=0
5797	156.240372	192.168.1.8	192.168.1.102	TLSv1.2	164	Application Data
5798	156.243687	192.168.1.102	192.168.1.8	TLSv1.2	164	Application Data
5799	156.287298	192.168.1.8	192.168.1.102	TCP	54	55705 → 8009 [ACK] Seq=3631 Ack=3635 Win=254 Len=0
5800	156.287479	192.168.1.8	192.168.1.102	TLSv1.2	164	Application Data
5801	156.291497	192.168.1.102	192.168.1.8	TLSv1.2	164	Application Data
5802	156.335153	192.168.1.8	192.168.1.102	TCP	54	54420 → 8009 [ACK] Seq=3521 Ack=3525 Win=255 Len=0
5803	156.494629	DASANNewtork_17:48:...	Broadcast	ARP	42	Who has 192.168.1.104? Tell 192.168.1.1
5804	156.531811	12:eb:b6:65:9f:ff	Broadcast	ARP	42	Who has 192.168.1.8? Tell 192.168.1.5
5805	156.531829	Intel_38:85:10	12:eb:b6:65:9f:ff	ARP	42	192.168.1.8 is at 8c:55:4a:38:85:10
5806	156.611419	150.171.28.11	192.168.1.8	TCP	58	443 → 51796 [RST, ACK] Seq=6945 Ack=2723 Win=0 Len=0
5807	157.494179	DASANNewtork_17:48:...	Broadcast	ARP	42	Who has 192.168.1.104? Tell 192.168.1.1
5808	158.349007	192.168.1.101	224.0.0.251	IGMPv2	46	Membership Report group 224.0.0.251
5809	158.494645	DASANNewtork_17:48:...	Broadcast	ARP	42	Who has 192.168.1.104? Tell 192.168.1.1
5810	160.662652	192.168.1.8	192.168.1.101	TLSv1.2	164	Application Data

.....0 = IG bit: Individual address (unicast)

Source: Intel_38:85:10 (8c:55:4a:38:85:10)
Type: IPv4 (0x0800)
[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.1.8, Dst: 192.168.1.101

0100 = Version: 4
..... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 150
Identification: 0x440f (17423)

010. = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0

0000	12 eb b6 65 9f ff 8c 55 4a 38 85 10 08 00 45 00	...e...U J8....E.
0010	00 96 44 0f 40 00 80 06 00 00 c0 a8 01 08 c0 a8	..D.@.....
0020	01 65 cc 18 1f 49 59 78 1f 9f d8 80 67 5d 50 18	.e...IYx ...g]P.
0030	00 fa 84 46 00 00 17 03 03 00 69 df 31 d1 a2 eb	...F....i.1...
0040	dc f3 ac 34 f0 89 70 99 04 b3 53 b9 84 b1 bb 49	...4..p...S....I
0050	2c 1a 3a f8 82 83 fa 37 4f 57 70 fe 28 41 ea 13	,.:...7 0Wp.(A..
0060	7d f1 13 7c 1e c7 8e 92 ce 9a 7b f8 8f b4 f1 87	}..-{-....
0070	1b 50 38 94 9c e9 fc 33 e2 ca 8c 1e 61 17 ba 35	..P8....3a...5
0080	90 74 21 31 1c 47 b1 25 c2 9c bf 92 8b 9d 01 08	..t!1-G.%
0090	b6 d6 65 5f 43 53 7d 73 13 90 ae f8 ba de 39 89	..e_CS}s9.
00a0	85 6e bb 28	..n.(

Considerations

- Wireshark cannot:
 - Break encryption
 - Hack networks
 - Capture traffic across a switch unless configured (SPAN, mirror, TAP)
- Wireshark can:
 - Decrypt some protocols if keys are provided
 - TLS using pre-master secrets
 - WPA2 Wi-Fi using passphrase + capture handshake



Wireshark cmd commands

S.No	Purpose	Command / Option	Example
1	Check TShark version	tshark -v	tshark -v
2	List network interfaces	tshark -D	tshark -D
3	Capture on an interface	tshark -i <id>	tshark -i 2
4	Stop capture manually	Press Ctrl + C	—
5	Capture only fixed number of packets	-c <count>	tshark -i 2 -c 100
6	Stop after time duration	-a duration:<sec>	tshark -i 2 -a duration:10
7	Stop after file reaches size	-a filesize:<kb>	tshark -i 2 -a filesize:100
8	Capture with filter (capture filter)	-f "<filter>"	tshark -i 2 -f "tcp port 80"
9	Display filter while capturing	-Y "<filter>"	tshark -i 2 -Y "dns"
10	Save capture to pcap file	-w <file>	tshark -i 2 -w capture.pcap
11	Limit packets + save to file	-c <count> -w <file>	tshark -i 2 -c 100 -w sample.pcap
12	Read pcap file	-r <file>	tshark -r capture.pcap
13	Read pcap with filter	-r <file> -Y "<filter>"	tshark -r cap.pcap -Y "http"
14	Show full packet details	-V	tshark -i 2 -V
15	Show detailed view for protocol	-O <protocol>	tshark -i 2 -O http
16	Export selected fields	-T fields -e <field>	tshark -r cap.pcap -T fields -e ip.src -e ip.dst
17	Write output to text file	> <filename>	tshark -r cap.pcap > output.txt
18	Capture only HTTP	-f "tcp port 80"	tshark -i 2 -f "tcp port 80"
19	Capture only DNS	-f "udp port 53"	tshark -i 2 -f "udp port 53"
20	Capture only DHCP	-f "port 67 or port 68"	tshark -i 2 -f "port 67 or port 68"
21	Capture only HTTPS	-f "tcp port 443"	tshark -i 2 -f "tcp port 443"



IP Addressing



IP Addressing

- **IP Addressing**

- Introduction to IP Addressing
- IPv4 Addressing
- Network Addressing
- Classless Inter-Domain Routing
- Private IP Address
- APIPA
- IPv6 Addressing
- Network Address Translation
- Subnet and Super net



Internet Protocol (IP) Address

- An Internet Protocol (IP) address is a **unique identifier** assigned to each device connected to a network. This address enables devices to locate each other and communicate across networks, such as local area networks (LANs) and the internet.
- An IP address functions like a "**home address**" for each device, directing data to the correct destination.
- There are two main types of IP addresses in use today:
 - **IPv4** (Internet Protocol version 4)
 - **IPv6** (Internet Protocol version 6)



Role of IP Addresses in Networking

- The IP address is vital for:
 1. **Routing data between networks:** Routers use IP addresses to forward packets to the correct destination.
 2. **Identifying devices in a network:** Each device on a network has a unique IP address, which is used to locate and communicate with other devices.
 3. **Providing security and control:** IP addresses allow network administrators to manage access, control routing, and set up security protocols.
- Each IP address consists of two parts:
 1. **Network portion:** Identifies the specific network.
 2. **Host portion:** Identifies the individual device within that network.



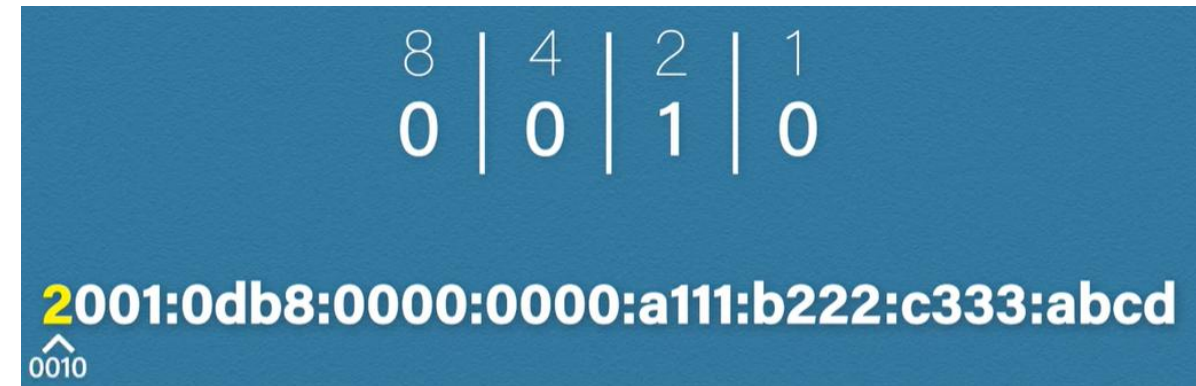
Public and Private

- IP addresses are also classified as PUBLIC or PRIVATE:
 - **Public IP Addresses** *are globally unique and accessible over the internet.* They are assigned by the Internet Assigned Numbers Authority (IANA) and are used to identify devices on a wide-scale network, such as the internet.
 - **Private IP Addresses** *are used within private networks (e.g., home or office networks).* These addresses are not unique globally and are meant for devices within a single network to communicate without directly connecting to the internet.



IPv6 Addresses

- IPv6 is of 128 bits.
- In IPv4 we have “Octet”, but here we have “Hextet”.
- IPv6 is hexa-decimal, it contains between (0 – 9) and (a-f)
 - Let’s say → 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
 - Actual value → 0 1 2 3 4 5 6 7 8 9 a b c d e f
- IPv6 uses “:” to separate IP addresses.



Key Features of IPv6

- Larger Address Space
- Simplified Header Structure
- No Need for NAT (Network Address Translation)
- Efficient Routing
- Support for Multicast
- Backwards Compatibility



IPv6 Address Types

- IPv6 addresses are also categorized by type, based on the intended application:
 - **Unicast:** Assigned to a single interface, allowing one-to-one communication.
 - **Multicast:** Intended for one-to-many communication.
 - **Anycast:** Assigned to multiple interfaces, with packets routed to the nearest device.
- IPv6 utilizes CIDR notation and prefixes to determine the address range:
 - **Global Unicast Addresses:** Range from 2000::/3, used for routing globally.
 - **Link-Local Addresses:** Range from fe80::/10, used within a single network segment.



CIDR (Classless Inter-Domain Routing)

- CIDR is a method for more flexible IP address allocation and efficient routing.
- It came as a replacement for the ***class-based IP addressing system*** (Classes A, B, C).
- CIDR allows for subnetting and aggregation of IP addresses, making it crucial for efficient network management and mitigating IPv4 address exhaustion.
- CIDR addresses use **variable-length subnet masking (VLSM)**, which allows network administrators to define IP addresses with varying subnet sizes, ensuring that IP addresses are allocated according to an organization's specific needs.



The Structure of CIDR Notation

- CIDR notation consists of an IP address followed by a slash (/) and a number that specifies the length of the network prefix.
- For example:
 - **192.168.1.0/24** means that the first 24 bits of the IP address are the network portion, and the remaining bits are available for host addresses.
 - **10.0.0.0/8** indicates that the first 8 bits are used for the network, allowing for a large range of host addresses within that network.



Benefits of CIDR

- **Efficient Use of IP Addresses:** CIDR allows for custom subnet sizes, ensuring that IP addresses are assigned based on need.
- **Reduced Routing Complexity:** Aggregating IP addresses into supernets minimizes routing table entries, improving routing efficiency.
- **Flexibility in Network Design:** Networks can be structured to match organizational requirements without the constraints of traditional IP classes.
- **Support for IPv6:** CIDR seamlessly integrates with IPv6, providing a robust method for handling extensive address spaces.



Common CIDR Notations and Their Subnet Sizes

CIDR Notation	Subnet Mask	Total Addresses	Usable Addresses
/8	255.0.0.0	16,777,216	16,777,214
/16	255.255.0.0	65,536	65,534
/24	255.255.255.0	256	254
/26	255.255.255.192	64	62
/30	255.255.255.252	4	2

Classful vs. Classless Addressing

- **Classful Addressing:** IP classes (A, B, and C) are considered classful as they follow predefined boundaries based on the first few bits of the address. While they simplify network design, they also limit flexibility and can lead to inefficient IP usage, as each class has a fixed number of hosts and network allocation.
- **Classless Inter-Domain Routing (CIDR):** CIDR was introduced to address the limitations of classful addressing. It allows networks to be defined by the subnet mask instead of the class, which conserves IP addresses by allocating them based on network needs rather than rigid class boundaries.



What is a Public IP Address?

- A public IP address is a globally unique IP address assigned to devices for communication over the internet.
- These addresses are accessible from anywhere on the internet, meaning devices with public IPs can send and receive data from other public IP addresses worldwide.
- Public IP addresses are issued by Internet Service Providers (ISPs) and are required for devices that need direct, unrestricted internet access, such as web servers or routers in home networks.



Public IP Addresses

- Globally Unique
- Internet-Routable
- Assigned by ISPs
- Examples:
 - Web Servers
 - Routers in home networks.

What is a Private IP Address?

- Private IP addresses are reserved for local network use only.
- They are not accessible from the public internet and are used within private networks like home, office, and corporate networks.
- Devices with private IP addresses can communicate with each other within the same network, but they cannot directly interact with devices on the internet without using a router or a Network Address Translation (NAT) service.



Private IP Address

- Local Use Only
- Reuse Across Networks
- NAT Required for Internet Access
- The IPv4 private address ranges are:
 - **Class A:** 10.0.0.0 to 10.255.255.255
 - **Class B:** 172.16.0.0 to 172.31.255.255
 - **Class C:** 192.168.0.0 to 192.168.255.255



Network Address Translation (NAT)

- *NAT is a technology that enables devices with private IPs to communicate over the internet by translating private IP addresses to a public IP address.*
- Routers use NAT to map multiple private IP addresses to a single public IP, which conserves IP addresses and allows entire networks to share a single public IP for internet access.
- Types of NAT:
 - **Static NAT:** Maps a single private IP to a single public IP, often used for hosting services.
 - **Dynamic NAT:** Maps private IPs to a pool of public IPs dynamically.
 - **Port Address Translation (PAT):** Maps multiple private IPs to a single public IP using different ports, commonly used in home networks.



Overview of IP Address Classes

Class	Range of First Octet	Default Subnet Mask	Number of Networks	Number of Hosts per Network	Usage
A	0 – 127	255.0.0.0	128	16,777,214	Large networks
B	128 – 191	255.255.0.0	16,384	65,534	Medium-sized networks
C	192 – 223	255.255.255.0	2,097,152	254	Small networks
D	224 – 239	Not applicable	Not applicable	Not applicable	Multicasting
E	240 – 255	Not applicable	Not applicable	Not applicable	Experimental and reserved

Subnetting

- **Subnetting** is the process of dividing a large IP network (network ID) into multiple smaller, more manageable logical subnetworks (subnets).
- It allows network administrators to optimize IP address usage, improve routing efficiency, and enhance security and broadcast containment.

- **Purpose of Subnetting**

- Efficient use of IP addresses
- Reducing network congestion
- Better network management
- Isolating broadcast domains
- Enhancing security and segmentation

- **Benefits**

- Reduces broadcast traffic
- Supports hierarchical addressing
- Improves routing scalability
- Enables network segmentation (VLANs, departments, zones)



Supernetting

- **Supernetting**, also called **route aggregation** or **network summarization**, is the process of combining multiple smaller contiguous networks into a single larger network address (supernet). It is the opposite of subnetting.
- Supernetting is commonly used in routing to reduce routing table size.
- **Purpose of Supernetting**
 - Reduce routing table entries
 - Simplify route advertisement
 - Improve router performance
 - Efficient IP route management in ISPs and large networks
- **Benefits**
 - Decreases routing overhead
 - Supports scalable hierarchical routing
 - Reduces bandwidth used for route updates
 - Ideal for ISPs and backbone networks







That's all Folks!